

Test 36: Byte-level robustness

Verdict: PASS | Generated: 2026-09-15 09:08:55 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Mixed: $|r|$ under 256-level quantization + RNG entropy (statistical). Sample accumulated to 25 000 (pass 1) / 50 000 (pass 2) eigenvalues; verdict = multi-window average over 120 windows

METHOD: Mixed: $|r|$ under 256-level quantization + RNG entropy (statistical). Sample accumulated to 25 000 (pass 1) / 50 000 (pass 2) eigenvalues; verdict = multi-window average over 120 windows

WHY W (DISORDER) IS CAPPED AT W_max ? 1.0 ? disorder ? vortex motion ? statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k W / (r_{ik}^2 + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy $q \cdot W$) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

$q \cdot W \gg t$ (e.g. $W=4$, $q=0.3 \Rightarrow V \approx 1.2$ next to a core): the disorder

landscape dominates the vortex motion ? vortices pin/steer the

eigenstates, states localize on the potential relief, and r drifts

from GUE (0.5992) toward Poisson (0.3863) by an essentially random

amount. A strongly disordered, spinning vortex system can therefore

emit ALMOST ANY statistics ? the numbers depend on the effective

rotation speed (W/t) and on the twisting/vortex arrangement (N_v ,

layout, charges) of that realization, not on the AB physics.

$q \cdot W \ll t$ ($W \ll 1$): the Aharonov-Bohm phases dominate the hopping,

states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_}W, W_{\text{max}} = 1.0)$:

they certify the PHASE physics, not the disorder physics.

Sample: 55290 eigs (pass 2 (secondary), target 50000, reached=yes); $W_{\text{eff}}=1.00$.

Byte robustness verdict = multi-window $|r| = 0.0009$ (tol 0.0200, 120 windows); single-window $|r| = 0.0061$ (diagnostic).

Test 36: byte robust ? $r_{256}=0.5756$, $|r|_{\text{multi}}=0.0009$ (120 windows), $|r|_{1\text{win}}=0.0061$ (diag), $H=7.9991$ bits, $z=-0.234$, $n=55288$? PASS

(no plots queued by this pass ? physics series live in the HARDCORE pass-2 report: hc_verdict_table queues a sub-check verdict map there, and the deep-audit tests queue their own)

Result

Reproduce with: `julia ab_cloud_v23.jl --test 36 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Plots

Independent verification guide

FULL COMPUTATION LOG

[08:25:16.507] [+31107.536s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[08:29:46.125] [+31377.154s] t37 pass 2 (secondary) realization 1: central block 5529 eigs, accumulated 5529 (target 50000)

[08:29:47.423] [+31378.451s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[08:34:04.812] [+31635.840s] t37 pass 2 (secondary) realization 2: central block 5529 eigs, accumulated 11058 (target 50000)

[08:34:05.610] [+31636.639s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[08:38:39.717] [+31910.745s] t37 pass 2 (secondary) realization 3: central block 5529 eigs, accumulated 16587 (target 50000)

[08:38:40.698] [+31911.726s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[08:42:58.835] [+32169.863s] t37 pass 2 (secondary) realization 4: central block 5529 eigs, accumulated 22116 (target 50000)

[08:42:59.617] [+32170.645s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[08:47:25.728] [+32436.756s] t37 pass 2 (secondary) realization 5: central block 5529 eigs, accumulated 27645 (target 50000)

[08:47:26.636] [+32437.664s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[08:51:40.917] [+32691.946s] t37 pass 2 (secondary) realization 6: central block 5529 eigs, accumulated 33174 (target 50000)

[08:51:41.498] [+32692.526s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[08:56:01.607] [+32952.636s] t37 pass 2 (secondary) realization 7: central block 5529 eigs, accumulated 38703 (target 50000)

[08:56:02.214] [+32953.242s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[09:00:21.506] [+33212.534s] t37 pass 2 (secondary) realization 8: central block 5529 eigs, accumulated 44232 (target 50000)

[09:00:22.299] [+33213.327s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[09:04:35.974] [+33467.002s] t37 pass 2 (secondary) realization 9: central block 5529 eigs, accumulated 49761 (target 50000)

[09:04:36.638] [+33467.666s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[09:08:55.278] [+33726.306s] t37 pass 2 (secondary) realization 10: central block 5529 eigs, accumulated 55290 (target 50000)

[09:08:55.292] [+33726.320s] calc: ?r? = 0.387930 (n_r=55288, n_eigs=55290) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.307] [+33726.335s] calc: ?r? = 0.657053 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: ?r? = 0.643724 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: ?r? = 0.624209 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: ?r? = 0.568482 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: ?r? = 0.522107 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: ?r? = 0.535764 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: ?r? = 0.652837 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: ?r? = 0.595610 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: ?r? = 0.624374 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: ?r? = 0.604567 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: ?r? = 0.485897 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: ?r? = 0.693192 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: $r = 0.683830$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: $r = 0.528901$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: $r = 0.519134$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: $r = 0.492447$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: $r = 0.622341$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: $r = 0.617574$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: $r = 0.607611$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: $r = 0.545788$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: $r = 0.540762$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: $r = 0.384361$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.336s] calc: $r = 0.605531$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.527459$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.492429$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.640946$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.815854$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.467241$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.655008$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.651800$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.476787$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.522209$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.526679$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.487008$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.639582$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.663925$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.501332$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.462589$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.486558$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.308] [+33726.337s] calc: $r = 0.571701$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $r = 0.655437$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $r = 0.669788$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $r = 0.681240$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $r = 0.655458$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $r = 0.628425$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $r = 0.585196$ ($n_r=13$, $n_{eigs}=15$) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.563536 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.692005 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.726317 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.665739 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.612582 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.672288 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.537854 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.462356 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.634012 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.544264 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.407715 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.540566 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.458490 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.572290 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.612044 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.745189 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.606921 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.673025 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.399152 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2 = 0.553205$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.548260 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.521588 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.546769 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.533360 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2 = 0.501554$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2_r = 0.508977$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.610973 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.626084 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.550313 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.550351 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.642275 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.529751 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.548388 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.658126 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2 = 0.557107$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2 = 0.636281$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2 = 0.497804$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.684932 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2_r = 0.687670$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2_r = 0.570463$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.610529 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.596610 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.617638 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2 = 0.380672$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2 = 0.525082$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2 = 0.596661$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.578898 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.512743 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.628458 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.650209 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: ?r? = 0.535112 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2_r = 0.599149$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2_r = 0.454103$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2 = 0.592745$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.337s] calc: $\chi^2_r = 0.490758$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: $\chi^2 = 0.482994$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.451370 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.611679 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.530496 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.515716 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.547860 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.699975 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: $\chi^2 = 0.619194$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: $\chi^2 = 0.372409$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: $\chi^2 = 0.670355$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.708571 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.607556 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: $\chi^2 = 0.621021$ (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.662679 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.484236 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.642217 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.541220 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.602689 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.309] [+33726.338s] calc: ?r? = 0.479526 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

[09:08:55.310] [+33726.338s] calc: ?r? = 0.599213 (n_r=13, n_eigs=15) [refs: GUE=0.5996 Poisson=0.3863]

CONSOLE CAPTURE

??

TEST 36: Byte-level robustness ? quantization + RNG entropy

Verify ?r? survives 256-level quantization + RNG byte uniformity

??

WHY W (DISORDER) IS CAPPED AT W_max ? 1.0 ? disorder ? vortex motion ? statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = q_k q_k W / (r_{ik} N_{\nu} + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy $q \bullet W$) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

? $q \bullet W$? t (e.g. $W=4$, $q=0.3$? $V \approx 1.2$ next to a core): the disorder

landscape dominates the vortex motion ? vortices pin/steer the

eigenstates, states localize on the potential relief, and ?r? drifts

from GUE (0.5992) toward Poisson (0.3863) by an essentially random

amount. A strongly disordered, spinning vortex system can therefore

emit ALMOST ANY statistics ? the numbers depend on the effective

rotation speed (W/t) and on the twisting/vortex arrangement (N_{ν} ,

layout, charges) of that realization, not on the AB physics.

? $q \bullet W$? t ($W \approx 1$): the Aharonov-Bohm phases dominate the hopping,

states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_}W, W_{\text{max}} = 1.0)$:

they certify the PHASE physics, not the disorder physics.

Sample target: 50000 eigenvalues [pass 2 (secondary)] (cfg.ab37_n_eigs_pass2)

Disorder: $W = 1.00 = \min(\text{ab_}W=4.00, W_{\text{max}}=1.00)$? $q \bullet W = 1.00$ vs hopping $t = 1.0$? AB phases dominate (byte check tests the phase physics)

realization 1/5, continuing (below target): +5529 central eigs ? accumulated 5529 / 50000 (11%)

realization 2/5, continuing (below target): +5529 central eigs ? accumulated 11058 / 50000 (22%)

realization 3/5, continuing (below target): +5529 central eigs ? accumulated 16587 / 50000 (33%)

realization 4/5, continuing (below target): +5529 central eigs ? accumulated 22116 / 50000 (44%)

realization 5/5, continuing (below target): +5529 central eigs ? accumulated 27645 / 50000 (55%)

realization 6/6, continuing (below target): +5529 central eigs ? accumulated 33174 / 50000 (66%)

realization 7/7, continuing (below target): +5529 central eigs ? accumulated 38703 / 50000 (77%)

realization 8/8, continuing (below target): +5529 central eigs ? accumulated 44232 / 50000 (88%)

realization 9/9, continuing (below target): +5529 central eigs ? accumulated 49761 / 50000 (100%)

realization 10/10: +5529 central eigs ? accumulated 55290 / 50000 (111%)

Raw $r = 0.3879$ (n=55288 spacings)

Universality class: Poisson (d_GUE=0.2117, d_GOE=0.1428, d_GSE=0.2883, d_Poi=0.0016)

(local window: n=15 central eigs, raw $r_{\text{local}} = 0.6571$)

(a) Byte-quantization robustness (256 levels):

legacy single window (diagnostic): $r_{\text{byte}} = 0.6631$, $r_{\text{local}} = 0.6571$, $|r| = 0.0061$

multi-window average (decisive; 120 windows $\times$ 15 eigs):

$r_{\text{raw}} = 0.5765$, $r_{\text{byte}} = 0.5756$, $|r| = 0.0009$ (tol=0.0200) ? PASS

(b) RNG byte-entropy / uniformity:

Shannon entropy = 7.9991 bits (target ? 7.9)

z (CLT approx, df=255) = -0.2342 ($|z| < 3.0$) ? PASS

Test 36: byte robust ? $r_{256} = 0.5756$, $|r|_{\text{multi}} = 0.0009$ (120 windows), $|r|_{1\text{win}} = 0.0061$ (diag), H=7.9991 bits, $z = -0.234$, n=55288 ? PASS